

Provenance-Preserving Evidence-Gated Knowledge-Graph Augmentation for Auditable Entity–Relation Extraction

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Abstract

Introduction: Graph-augmented extraction can retrieve useful associations, but a retrieved association is not necessarily a fact stated in the current sentence. Fluent generators also make it difficult to trace a predicted relation to its source span and to audit why it was accepted. This gap is consequential when extracted assertions are used in safety-related knowledge systems.

Methods: We develop a provenance-preserving, evidence-gated framework that separates graph-informed candidate generation from output acceptance. A graph derived only from training annotations supplies bounded concept and relation hints to QLoRA-adapted Qwen3-4B extractors. Deterministic span reconstruction, entity acceptance, and relation verification then retain only source-linked, type-compatible assertions. We evaluate the framework on ADE and CoNLL04 using a shared strict character-span protocol and report the underlying dataset distributions.

Results: The completed test comparisons show that the provenance-gated system improves relation F1 over its matched source-only generator on both benchmarks and records explicit acceptance decisions. The relation improvement is retained for the reported CoNLL04 sensitivity analysis after exact train–test text overlaps are excluded.

Discussion: The results support controlled graph augmentation as an auditable extraction layer for human review. The gates establish structural admissibility and source linkage, not semantic truth, clinical causality, or universal accuracy dominance; the strongest supervised baseline remains competitive in overall extraction F1.

Keywords: entity and relation extraction, knowledge graph, provenance, evidence grounding, adverse drug effects, selective prediction

1. Introduction

Structured information extraction connects unstructured text to searchable entities, typed relations, and downstream knowledge systems. In applications related to safety, an extracted association must also be inspectable: users need to know which mention was identified, which sentence supported the relation, and which transformations produced the final assertion. Drug-related adverse-effect extraction illustrates this requirement directly. A model that recognizes a drug and an adverse effect but invents an association between them can create a misleading record even when its output is syntactically valid. General-domain extraction provides a complementary test of the same technical issue across a broader set of entity and relation labels.

Neural entity and relation models provide effective approaches to the recognition problem [1, 2]. Generative models offer a flexible alternative in which a common structured output format can be reused across label inventories. However, fluent generation does not guarantee that the predicted objects are supported by their input [3]. This distinction matters when extracted triples are inserted into a graph: errors become available to later retrieval or reasoning stages, potentially propagating beyond the sentence that generated them.

Knowledge graphs can contribute typed domain priors [4], and retrieval augmentation can expose those priors without requiring all knowledge to be stored in model parameters [5]. Their use introduces a specific authority problem. A relation observed in a training sentence may suggest what to search for, but it does not establish that the same relation holds in a new sentence. A training graph may also return an example’s own annotation or reveal content shared across data splits. The useful question is therefore not only whether graph context increases recall, but whether its eligibility and influence can be controlled and audited.

We address this problem with a staged architecture. Separate source-only, exact-anchor, and high-recall graph-conditioned generators establish a matched comparison of input conditions. A deterministic relation verifier checks structural and source-evidence requirements. A complementary entity gate uses agreement, source-matched anchors, or verified relation endpoints to accept a subset of the graph-conditioned candidates. The final system combines the accepted entities with verified relations whose endpoints survive. The architecture uses existing representation and adaptation methods;

its contribution is their governed composition, not a new backbone or a new optimization algorithm.

This paper concentrates on ADE [6] and CoNLL04 [7]. ADE supplies a drug-safety-related extraction task with two entity labels and one relation label. CoNLL04 supplies a general-domain control with four entity labels and five directed relation labels. Both use English sentence-level inputs, making it possible to examine the extraction and verification mechanisms without importing a long-document windowing problem. They are trained and evaluated separately; this is not a cross-dataset zero-shot transfer experiment.

The contributions are threefold:

- A provenance-aware graph-context policy that uses only training annotations, excludes candidates supported only by the current example, and clears graph hints for exact train/evaluation text overlaps.
- A deterministic composition of source-span reconstruction, entity selection, and relation verification that retains evidence-linked assertions and records the reasons behind output decisions.
- A reproducible two-benchmark analysis that combines split and label statistics, uniform strict-span scoring, validation ablations, completed test comparisons, and an exact-overlap sensitivity analysis.

The intended contribution is an auditable extraction layer for analyst-facing knowledge systems. It does not convert benchmark relation labels into clinical causality, establish calibrated risk probabilities, or validate autonomous safety decisions.

2. Related work

2.1. Joint entity and relation extraction

Entity recognition must resolve both mention boundaries and types; relation extraction additionally depends on recovering the correct directed endpoints [1]. Global consistency constraints have a long history in this setting, including the formulation associated with the CoNLL04 benchmark [7]. SpERT enumerates spans and jointly models entities and relations using a pretrained transformer [2]. It is a particularly relevant comparison because it performs the same joint task without requiring a generative JSON interface. Our experiments retrain SpERT from a clean base encoder on the same

71 training partitions rather than using a released task checkpoint that may
72 already include development data.

73 GLiNER and GLiREL provide label-conditioned alternatives for entity
74 recognition and relation classification [8, 9]. These models differ from locally
75 fine-tuned generators in supervision, pretraining, and output construction.
76 We therefore report the implemented train-calibrated pipeline as a named
77 external baseline, not as an exhaustive assessment of either model family.
78 A same-backbone zero-shot generator further separates the benefit of local
79 adaptation from that of graph conditioning.

80 2.2. Graph context and constrained generation

81 Knowledge graphs represent typed entities and relations explicitly, while
82 retrieval augmentation uses selected records as model context [4, 5]. GenIE
83 constrains generative information extraction with a structured output space
84 and entity/relation catalogues [10]. Catalogue validity and source support
85 are different requirements: an admissible entity or relation may still not be
86 asserted in the current input. Our graph context is consequently treated as
87 a candidate prior whose eligibility depends on training provenance and local
88 source matching.

89 BGE-M3 supplies frozen text representations for semantic pattern re-
90 trieval [11]. It is neither a graph neural network nor a jointly optimized
91 component of the extractor. QLoRA enables compact adaptation of the
92 Qwen3 backbone [12, 13]. Neither method alone defines whether a candi-
93 date should enter the final assertion graph; the proposed gates make that
94 acceptance policy explicit.

95 2.3. Evidence and safety-oriented knowledge construction

96 Hallucination research distinguishes well-formed outputs from source-
97 faithful ones [3]. In safety-related graph construction, prompt-based pro-
98 cessing of accident reports likewise illustrates the need to trace generated
99 knowledge back to its sources [14]. ADE offers a more controlled benchmark
100 for a drug-safety-related relation task [6]. Nevertheless, an ADE label is
101 an annotated textual association, not a patient-level causal conclusion. The
102 present study evaluates extraction and implemented evidence constraints,
103 rather than the medical or operational validity of downstream decisions.

104 3. Datasets and distribution analysis

105 3.1. Benchmark sources and partitions

106 We use the sentence-level ADE and CoNLL04 files distributed in SpERT
 107 format. Counts refer to that preprocessed distribution rather than every
 108 sentence in the original source corpora. CoNLL04 retains its supplied train-
 109 ing, development, and test partitions. The ADE files contain 3,845 training
 110 sentences and 427 test sentences; a deterministic development slice of 384
 111 sentences is removed from the supplied training pool, leaving 3,461 train-
 112 ing sentences. The slice is selected by sorting SHA-256 hashes of the string
 113 formed from the fixed split seed `public-full-42` and each original record
 114 identifier. It is not selected by validation performance. This is one fixed ADE
 115 split, not an average over the multiple folds sometimes used in the literature.

116 Table 1 gives counts recomputed from the exact imported artifacts. The
 117 unit called a document in the software is one benchmark sentence, not a whole
 118 medical case report or news article. Both datasets are English; the results
 119 therefore support neither multilingual nor cross-lingual claims. Published
 120 annotations are imported rather than locally re-annotated.

Table 1: Dataset inventory for the exact experimental partitions. Length is the number of whitespace-delimited tokens in the reconstructed source text.

Dataset	Split	Sentences	Entities	Relations	Mean length
ADE	train	3461	8717	5470	21.17
ADE	validation	384	996	627	21.82
ADE	test	427	1126	724	21.24
CoNLL04	train	922	3377	1283	28.77
CoNLL04	validation	231	893	343	30.27
CoNLL04	test	288	1079	422	28.94

121 3.2. Entity and relation distributions

122 ADE contains `Drug` and `Adverse-Effect` entities and a single
 123 `Adverse-Effect` relation label. The imported endpoint convention is
 124 *adverse effect to drug*; we preserve it rather than reversing it to a more
 125 intuitive drug-to-effect direction. CoNLL04 contains `Loc`, `Org`, `Peop`,
 126 and `Other` entities, with `Work_For`, `Kill`, `OrgBased_In`, `Live_In`, and
 127 `Located_In` relations.

128 The training-label distributions are summarized in the accompanying ta-
 129 bles, and Table 2 lists exact label counts for every partition. The single
 130 ADE relation label eliminates inter-class relation confusion but does not
 131 eliminate endpoint detection, direction, or false-pair errors. CoNLL04 in-
 132 troduces a broader relation inventory and more heterogeneous entity roles.
 133 Consequently, a higher aggregate F1 on ADE should not be interpreted as
 134 evidence that drug-safety reasoning is inherently easier than general-domain
 135 reasoning; the supervised extraction tasks differ in structure and size.

Table 2: Exact entity and relation annotation counts in the imported files.

Dataset	Kind	Label	Train	Validation	Test
ADE	entity	Adverse-Effect	4631	529	616
ADE	entity	Drug	4086	467	510
ADE	relation	Adverse-Effect	5470	627	724
CoNLL04	entity	Loc	1219	322	427
CoNLL04	entity	Org	616	170	198
CoNLL04	entity	Other	455	118	133
CoNLL04	entity	Peop	1087	283	321
CoNLL04	relation	Kill	179	42	47
CoNLL04	relation	Live_In	330	91	100
CoNLL04	relation	Located_In	247	65	94
CoNLL04	relation	OrgBased_In	271	76	105
CoNLL04	relation	Work_For	256	69	76

136 3.3. Sentence lengths and coverage

137 The mean training lengths are 21.17 reconstructed tokens for ADE and
 138 28.77 for CoNLL04. Their test means are 21.24 and 28.94, respectively.
 139 The cumulative distributions describe all three partitions rather than only
 140 the training set. These lengths are descriptive whitespace-token counts, not
 141 Qwen tokenizer counts. The largest observed reconstructed length is 97 for
 142 ADE and 118 for CoNLL04 across the imported partitions.

143 Every imported sentence contains at least one annotated relation. This is
 144 an important property of the selected benchmark files: they do not measure
 145 false-positive behavior over a deployment stream dominated by sentences
 146 with no relevant relations. We retain the supplied benchmark composition
 147 and do not create additional negative sentences for the reported tests.

148 3.4. Split overlap and graph-context isolation

149 The graph is built separately for each dataset from its training annotations only. An exact-overlap check compares case-folded source strings after
150 collapsing whitespace. ADE has no train/development or train/test exact-
151 text overlap under this check. CoNLL04 has 14 development sentences and
152 19 test sentences whose text matches training content. For these sentences,
153 both exact-anchor and high-recall graph context are cleared before genera-
154 tion.
155

156 Clearing graph hints does not undo exposure of the trained generator to
157 duplicate training text. We therefore retain the supplied partitions for the
158 main comparison and report an additional fixed-prediction test analysis that
159 excludes the 19 overlapping CoNLL04 test sentences. We do not claim that
160 the benchmark is duplicate-cluster isolated or that this exact-match check
161 detects all near duplicates.

162 4. Proposed method

163 4.1. Source-grounded representation

164 Let a sentence x be the deterministic single-space concatenation of its
165 released tokens. An entity mention is represented as

$$e = (b, u, y), \quad x[b : u] = m, \quad (1)$$

166 where $[b, u)$ is a character interval, y is an entity label, and m is the copied
167 mention text. A relation $r = (e_s, z, e_t)$ has typed directed endpoints and
168 relation label z . Source identifiers and evidence spans are stored with these
169 objects. Repeated strings at different offsets remain distinct mentions. The
170 character-span requirement in Equation (1) prevents a normalized name
171 alone from receiving credit for the wrong occurrence.

172 The imported benchmark assigns every relation the status `explicit` and
173 uses its source sentence as the evidence span. This is a representation conven-
174 tion, not a new annotation of minimal causal evidence or uncertainty. Since
175 both endpoints occur in the same input sentence, evidence co-occurrence is
176 a relatively weak condition in these experiments.

177 4.2. Architecture and comparison systems

178 Figure 1 summarizes the implemented flow. SOE (Source-Only Extrac-
179 tion), EAE (Exact-Anchor Extraction), and HRGE (High-Recall Graph Ex-
180 traction) are separately trained QLoRA adapters on the same Qwen3-4B

181 snapshot. SOE uses source text and the training-derived ontology without
 182 graph hints. EAE receives conservative concept hints, while HRGE receives
 183 bounded anchors, edge priors, and semantic relation patterns.

184 EVGE (Evidence-Verified Graph Extraction) applies the relation verifier
 185 to HRGE. CFE (Conservative-Fusion Extraction) pairs accepted HRGE en-
 186 tities with endpoint-filtered raw HRGE relations. PGE (Provenance-Gated
 187 Extraction) pairs the same accepted entities with endpoint-filtered verified
 188 relations. These three systems are deterministic post-processing variants,
 189 not additional trained networks. In deployment PGE needs EAE and HRGE
 190 candidates; SOE is a comparison branch and does not participate in fusion.

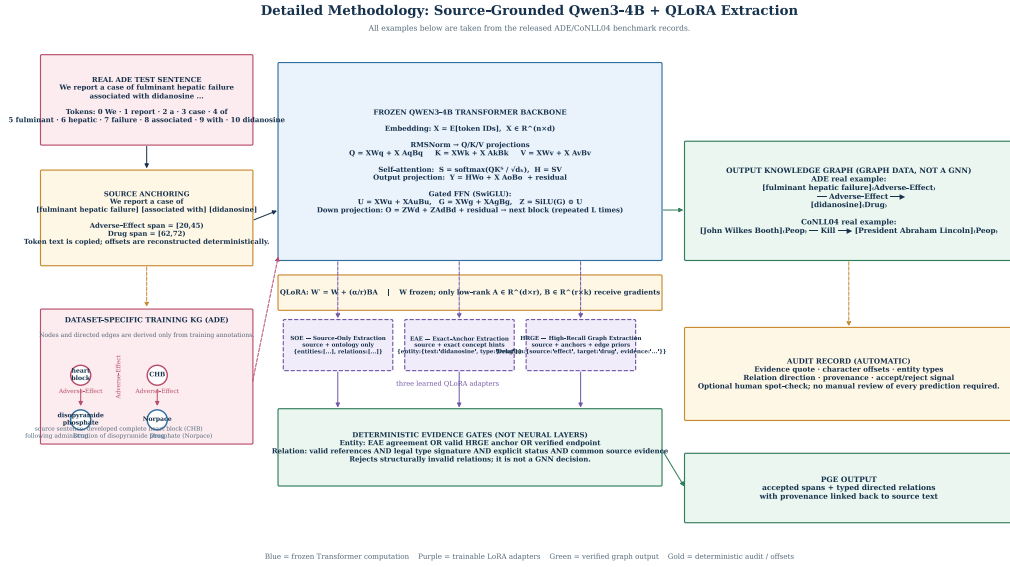


Figure 1: Detailed methodology for source-grounded Qwen3-4B extraction with QLoRA adaptation, dataset-specific training graphs, deterministic evidence gates, provenance-preserving graph output, and automatic audit records.

191 4.3. Provenance-controlled graph context

192 Each training graph stores mention strings, types, directed relations, and
 193 the sentence identifiers supporting them. For a training sentence d , a candi-
 194 date is excluded if its only supporting provenance is d . Candidates with other
 195 supporting sentences remain eligible; their stored aggregate support statistics
 196 are not recomputed as a fully leave-one-out graph. This distinction prevents

197 interpreting the provenance policy as complete removal of every contribution
 198 from the current example.

199 EAE uses at most 20 exact-matching concept hints with type balancing.
 200 HRGE uses three context channels. First, source-matched anchors must meet
 201 an empirical type-purity threshold of 0.8 and an English string-length thresh-
 202 old of four characters. Second, a typed edge prior requires support outside
 203 the current example and both endpoint strings in the current source. Third,
 204 frozen BGE-M3 retrieves semantically related relation-pattern records. If
 205 $\mathbf{H}$ contains normalized eligible record embeddings and $\mathbf{q}$ is the normalized
 206 source embedding, retrieval scores are

$$\mathbf{s} = \mathbf{H}\mathbf{q}, \quad \mathcal{I} = \text{TopK}_k\{i : s_i \geq \tau\}. \quad (2)$$

207 Equation (2) ranks candidate context, not accepted facts. The frozen HRGE
 208 limits are 12 anchors with at most two per type, six edge priors, and four
 209 semantic patterns with cosine similarity at least 0.72. The prompt explicitly
 210 identifies these objects as hints. Exact-overlap quarantine overrides retrieval
 211 and clears all graph context for the affected evaluation sentences.

212 4.4. Compact generation and deterministic reconstruction

213 The generator emits entity identifiers, copied text and types, plus relation
 214 identifiers, directed endpoint references, types, and explicit status. It is not
 215 asked to calculate source offsets. Deterministic expansion assigns copied
 216 mentions to occurrences in the supplied sentence and reconstructs character
 217 spans; unmatched strings and invalid references are recorded. This separates
 218 language-model candidate generation from source-coordinate bookkeeping.

219 All three generators use one training epoch, batch size one, gradient ac-
 220 cumulation four, and learning rate 2×10^{-4} . QLoRA uses rank eight, alpha
 221 16, dropout 0.05, NF4 double quantization, and bfloat16 computation. At-
 222 tention query/key/value/output projections and feed-forward gate/up/down
 223 projections are adapted; the backbone remains frozen. AdamW uses the Py-
 224 Torch defaults for its other parameters, without a learning-rate scheduler or
 225 warm-up. The maximum training sequence is 4,096 tokens, and prompt to-
 226 kens are masked from the loss. Training uses the same dataset partition for
 227 all three branches and seed 42 throughout the primary comparisons.

228 Inference uses the recorded 4,096-token input limit, 1,024-token genera-
 229 tion budget, and 90-second job limit, with two inference workers on the local
 230 GPU. The backbone revision and all input hashes are recorded in the artifact

manifest. Failed jobs are represented by empty predictions, not removed from the denominator. These experiments use the full supplied sentence, without annotation-guided rescue windows or label-dependent sentence truncation.

4.5. Relation verification and entity acceptance

For each relation label z , training annotations define permitted source-type and target-type sets S_z and T_z . The implemented type constraint is $y_s \in S_z$ and $y_t \in T_z$; it does not additionally require that every specific source/target type pair has been observed together. The verifier $V(r)$ also requires existing distinct entity references, a known relation label, explicit status, valid evidence, and both endpoint strings in a common source-evidence span. Its accepted set is

$$R_V = \{r \in R_H : V(r) = 1\}. \quad (3)$$

The entity gate uses three Boolean signals:

$$A(e) = A_{\text{agree}}(e) \vee A_{\text{anchor}}(e) \vee A_{\text{endpoint}}(e), \quad E_A = \{e \in E_H : A(e) = 1\}. \quad (4)$$

Agreement means that EAE predicts the same case-folded, whitespace-normalized text and type. Anchor support requires a same-type source-gated HRGE anchor. Endpoint support means that the entity participates in a relation retained by Equation (3). The three signals in Equation (4) are complementary but not statistically independent; verified endpoints are a rule-based signal, not a second human judgement. Duplicate normalized text/type candidates are collapsed within the sentence.

Let $F(R, E)$ retain relations in R only when both referenced entities survive in E . The two compositions are

$$G_{\text{CFE}} = (E_A, F(R_H, E_A)), \quad G_{\text{PGE}} = (E_A, F(R_V, E_A)). \quad (5)$$

Equation (5) makes the difference between raw-relation fusion and verified fusion explicit. Accept/reject decisions are logged. A retained relation satisfies the stated rules, but the rules alone cannot establish whether the sentence asserts the proposed meaning.

5. Experimental protocol

5.1. Research questions and scope

We ask whether PGE improves strict entity and relation extraction over SOE; how graph context and deterministic selection contribute on development data; and whether observed gains persist after exact-overlap sentences

are removed. External baselines establish whether the generative design is competitive with other extraction families, rather than assuming that improvements over SOE imply state-of-the-art performance.

The two-benchmark reporting scope was narrowed retrospectively after validation and available test results had been inspected. It is a focused analysis, not a preregistered or representative sample of all attempted extraction domains. Within the existing protocol, model and gate settings were frozen before test release. SOE and PGE were promoted when validation relation F1 did not regress by more than one percentage point and the required completion and evidence checks passed. This was a pragmatic point-estimate promotion rule, not a statistical non-inferiority test. The current paper uses the completed seed-42 tests and does not infer multi-seed stability from them.

5.2. Baselines and evaluation parity

The internal development comparison includes all six named systems. The test table includes the two promoted systems and three completed external comparisons. Qwen3-4B zero-shot uses the same backbone without local adapters, followed by the recorded source expansion and relation verifier. SpERT is trained from `bert-base-cased` on the same training partitions for 20 epochs, with batch size two and a relation threshold of 0.4. Its final checkpoint is used without selecting a test-best epoch. GLiNER+GLiREL uses predicted, not gold, entities and relation calibration confined to an inner training subset; the entity threshold is 0.5. These are heterogeneous baseline configurations, not equal-compute experiments.

The recorded GLiREL calibration selects canonical relation labels and a relation threshold of 0.0 for ADE, and naturalized labels with threshold 0.1 for CoNLL04. No development or test gold is used to choose those settings.

All reported main-table predictions are rescored with one public character-span evaluator. An entity key contains its exact source interval and type. A relation key contains both typed directed entity keys and the relation label. Predicted evidence offsets must resolve to the supplied source; the evaluator does not recover missing offsets by searching normalized text. Duplicate keys receive one vote, invalid objects remain unmatched predictions, and omitted jobs remain empty. For pooled counts,

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2TP}{2TP + FP + FN}. \quad (6)$$

Equation (6) is applied across sentences within each dataset, not after averaging their individual F1 scores. Because all imported statuses are explicit, claim-status-aware scores do not form a separate uncertainty benchmark. The unified evaluator removes small denominator differences caused by legacy duplicate handling; unchanged source artifacts and a reconciliation report preserve the earlier metrics.

5.3. Statistical and compliance analyses

We compute 20,000 paired bootstrap resamples of test sentences, using seed 20260830 and fixed predictions, and recompute pooled F1 within each resample. The percentile interval describes sentence-sampling uncertainty for the completed model pair, not variability across training seeds. Intervals are descriptive, unadjusted for multiple comparisons, and are not used to change the frozen models or gates.

Evidence coverage and rule-invalid counts are measured separately from extraction F1. These compliance counts refer to emitted prediction objects before the evaluator’s key deduplication. Exact-overlap sensitivity removes the same affected test sentence identifiers from both systems, without retraining or regenerating predictions. Test labels are used for reporting statistics and scoring only, not for constructing graph hints or choosing thresholds in this revision.

6. Results

6.1. Completed test comparisons

Table 3 reports the completed test results under the common evaluator. PGE raises ADE relation precision from 80.62% to 87.00% and recall from 67.82% to 72.10%, giving relation F1 of 78.85% compared with 73.67% for SOE. Entity F1 remains essentially unchanged, at 87.93% versus 87.97%. Thus, the main improvement is relational rather than a general gain in mention recognition.

On CoNLL04, PGE raises relation precision from 54.61% to 62.40% and recall from 56.16% to 57.82%. Relation F1 increases from 55.37% to 60.02%, while entity F1 decreases from 83.83% to 83.20%. The composition trades some entity coverage for a better relation precision–recall balance in this run.

SpERT has the highest entity and relation F1 among these completed systems: 91.37%/84.08% on ADE and 89.02%/69.72% on CoNLL04. PGE therefore improves the matched generative baseline but does not surpass this

Table 3: Completed test comparisons, strict source-character-span micro scores (%). Qwen3-ZS is Qwen3-4B zero-shot with the recorded verifier; GLiNER+GLiREL uses training-only calibration. Each dataset uses its own test partition. Locally trained systems use seed 42; all rows are rescored with the same evaluator.

Dataset	System	Ent. P	Ent. R	Ent. F1	Rel. P	Rel. R	Rel. F1
ADE	Qwen3-ZS + verifier	63.83	63.32	63.58	39.36	22.24	28.42
ADE	GLiNER + GLiREL	79.30	64.65	71.23	60.31	43.23	50.36
ADE	SpERT	90.65	92.10	91.37	82.19	86.05	84.08
ADE	SOE	92.21	84.10	87.97	80.62	67.82	73.67
ADE	PGE	92.89	83.48	87.93	87.00	72.10	78.85
CoNLL04	Qwen3-ZS + verifier	48.78	53.94	51.23	30.92	19.19	23.68
CoNLL04	GLiNER + GLiREL	60.78	14.37	23.24	13.79	3.79	5.95
CoNLL04	SpERT	88.61	89.43	89.02	71.01	68.48	69.72
CoNLL04	SOE	85.83	81.93	83.83	54.61	56.16	55.37
CoNLL04	PGE	86.04	80.54	83.20	62.40	57.82	60.02

trained span-based model. PGE also exceeds the recorded zero-shot and GLiNER+GLiREL pipelines on both extraction tasks. The latter observation is specific to their implemented label mapping and calibration; it is not a claim that every configuration of those model families would perform worse.

6.2. Paired differences and uncertainty

Using unrounded counts, PGE-minus-SOE relation F1 is 5.18 percentage points on ADE, with a paired 95% interval of [0.31, 10.29], and 4.65 points on CoNLL04, with an interval of [0.04, 9.12]. The positive point estimates are consistent with the test-table improvements, but the intervals are broad and their lower bounds are close to zero. We interpret them as evidence about these completed test predictions, not as a general guarantee across seeds. Entity-F1 differences are -0.04 points on ADE, interval $[-1.75, 1.62]$, and -0.63 on CoNLL04, interval $[-2.38, 1.04]$; neither supports an entity-F1 improvement claim.

6.3. Development-set component analysis

Table 4 separates candidate generation from deterministic selection on development data. On ADE, HRGE has relation F1 of 76.29%, above SOE’s 73.16%. EVGE is identical to HRGE, and PGE is identical to CFE, showing that the relation verifier removes no additional relations in these variants. The entity gate reduces HRGE recall and gives PGE relation F1 of 75.91%. It is therefore incorrect to attribute all ADE gains to verification or to claim that every stage independently improves F1.

On CoNLL04, HRGE relation F1 is 58.26%, below SOE’s 60.24%. Verification raises it to 59.60%, and the final composition reaches 60.19%. This is effectively unchanged from SOE at the development point estimate, rather than a clear development-set gain. Entity precision rises after gating, while recall and F1 decrease. The distinction between development and test behavior is retained rather than replacing the ablation with a test-selected variant.

Table 4: Development-set ablation under the common strict-span evaluator (micro %, seed 42). These rows characterize the frozen architecture and are not additional promoted test systems.

Dataset	System	Ent. P	Ent. R	Ent. F1	Rel. P	Rel. R	Rel. F1
ADE	SOE	89.25	82.53	85.76	78.82	68.26	73.16
ADE	EAE	90.49	81.22	85.61	80.84	67.30	73.46
ADE	HRGE	90.35	85.54	87.88	80.04	72.89	76.29
ADE	EVGE	90.35	85.54	87.88	80.04	72.89	76.29
ADE	CFE	90.48	83.94	87.08	81.39	71.13	75.91
ADE	PGE	90.48	83.94	87.08	81.39	71.13	75.91
CoNLL04	SOE	84.94	82.08	83.49	61.33	59.18	60.24
CoNLL04	EAE	83.71	82.87	83.29	58.10	60.64	59.34
CoNLL04	HRGE	82.35	81.52	81.94	60.06	56.56	58.26
CoNLL04	EVGE	82.35	81.52	81.94	62.99	56.56	59.60
CoNLL04	CFE	84.97	78.50	81.61	63.37	55.98	59.44
CoNLL04	PGE	84.97	78.50	81.61	65.08	55.98	60.19

6.4. Rule compliance and its interpretation

The test SOE output contains 31 rule-invalid relation objects among 610 ADE objects and 16 among 436 CoNLL04 objects. PGE retains 600 and 392 relation objects, respectively, with zero rule-invalid objects. HRGE already has zero invalid ADE relations and only three invalid CoNLL04 relations among 413 objects. These counts distinguish the contribution of candidate quality from that of the verifier.

The measured unsupported-evidence count is zero for SOE, HRGE, and PGE on both test datasets. Because each relation’s evidence is the whole sentence, co-occurrence is easy to satisfy once endpoints resolve. Thus, these experiments do not demonstrate a reduction from a nonzero unsupported-claim rate. They show explicit evidence binding and rule compliance, while the gold-referenced F1 scores measure a different property: whether the predicted typed relation agrees with the annotation.

371 6.5. *Exact-overlap sensitivity*

372 Removing the 19 exact train/test overlaps leaves 269 CoNLL04 test sen-
373 tences, 1,023 gold entities, and 403 gold relations. SOE relation F1 becomes
374 53.79% and PGE becomes 59.10%, a 5.31-point difference; entity F1 is 83.02%
375 and 82.90%, respectively. The observed relation advantage therefore persists
376 without those exact overlaps. This analysis addresses exact repeated text,
377 not all forms of semantic overlap or pretraining contamination. ADE has no
378 such overlaps, so its sensitivity result is identical to the full test.

379 7. Discussion

380 7.1. *What the two benchmarks establish*

381 The common result is a relation-F1 improvement over source-only gen-
382 eration, not universal superiority over extraction architectures. The gains
383 arise alongside higher relation precision on both tests, while entity F1 stays
384 similar or declines slightly. This is consistent with an architecture designed
385 to govern candidate acceptance rather than simply maximize output volume.
386 The validation ablation further shows that graph conditioning and selection
387 interact: recall-oriented context can help on ADE but does not automatically
388 improve CoNLL04, and a verifier may be inactive when candidates already
389 satisfy its rules.

390 The distribution analysis helps interpret these differences. ADE has a
391 larger training pool, shorter average sentences, and a single relation class.
392 CoNLL04 has fewer training sentences, longer inputs, and five relation labels.
393 These factors are confounded rather than experimentally isolated. Their
394 association with the observed scores motivates separate dataset reporting; it
395 does not establish a causal explanation for the performance gap.

396 7.2. *Auditability as a distinct design objective*

397 A system can be less accurate than SpERT while still exposing a use-
398 ful acceptance mechanism for generated graph objects. The contribution is
399 strongest when the entire path is inspectable: graph support identifies why
400 a candidate was suggested; reconstructed offsets identify where it appears;
401 gate records identify why it was retained or rejected. However, this benefit
402 is architectural. No user study in the present experiments measures reduced
403 analyst effort, improved decisions, or greater trust. It would be inappropriate
404 to substitute perfect rule compliance for such evidence.

405 The gates also impose a coverage cost. Agreement operates on normal-
406 ized text/type pairs, and repeated mentions may be collapsed during fusion
407 even though the evaluator distinguishes offsets. This can remove legitimate
408 occurrences. A legal endpoint signature and sentence-level co-occurrence are
409 insufficient to prove a relation’s meaning, and three gate signals should not be
410 interpreted as three independent confirmations. These observations explain
411 why the final output remains a candidate assertion graph for review.

412 7.3. *Implications for safety-related applications*

413 ADE links the experiments to drug-safety information extraction, while
414 CoNLL04 tests general relational structure. The framework can present an
415 analyst with the candidate relation, source sentence, typed endpoints, and
416 provenance of its acceptance path. An analyst can then accept, amend, or
417 reject the proposed association. Absence from the retained graph is not
418 evidence of the absence of an adverse effect or another real-world relation.

419 The reported task stops before pharmacovigilance judgement, patient-
420 specific causal attribution, or operational action. Neither dataset validates
421 risk propagation, resilience assessment, or autonomous intervention. Ex-
422 tending this work toward those applications requires task-specific outcome
423 measures and expert assessment in addition to extraction F1.

424 8. Limitations and responsible use

425 The focused, retrospectively narrowed dataset scope and single completed
426 training seed limit generalization. The fixed ADE partition is not directly
427 interchangeable with published multi-fold results. Both corpora are English
428 and consist of relation-bearing sentences; multilingual behavior, long-report
429 processing, and false-positive rates on unrestricted text streams are outside
430 the evaluated setting. Sentence-level bootstrap intervals cannot recover de-
431 pendence between sentences originating from the same upstream article when
432 those article identifiers are unavailable.

433 Exact-overlap quarantine prevents direct graph hints for known repeated
434 source text but does not remove the generator’s training exposure. The sen-
435 sitivity analysis excludes known exact duplicates only. Frozen pretrained en-
436 coders and generators may also have seen related material during pretraining,
437 which the available artifacts cannot rule out. The ontology’s allowed source
438 and target sets are less restrictive than a catalogue of observed joint type
439 pairs.

440 The one-epoch generative systems, 20-epoch SpERT, and pretrained
441 label-conditioned models do not form a compute-matched benchmark. PGE
442 requires two adapted generators plus frozen retrieval, so compact adaptation
443 must not be confused with measured end-to-end cost savings. No new
444 clinical annotations or independent evidence-judgement labels were collected
445 for this paper. Public availability of a dataset or model does not by itself
446 permit every downstream redistribution or commercial use; original licenses
447 remain applicable.

448 **9. Conclusions**

449 This study presents a provenance-preserving, evidence-gated graph-
450 augmentation framework and evaluates it on ADE and CoNLL04. Dataset
451 statistics expose the label and input-length differences underlying the two
452 benchmarks. Under a unified strict-span evaluator, the completed PGE
453 tests improve relation F1 over SOE by 5.18 and 4.65 percentage points,
454 respectively, while offering an explicit source-linked acceptance path. The
455 CoNLL04 relation advantage remains after exact train/test overlaps are
456 excluded. SpERT remains stronger in overall extraction F1, and the
457 development ablations show that the gates are not uniformly beneficial in
458 isolation. The supported outcome is controlled, auditable graph-informed
459 extraction for human review, rather than a claim of universal accuracy
460 dominance or validated autonomous safety reasoning.

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464 **Declaration of competing interest**

465 The authors declare that they have no known competing financial in-
466 terests or personal relationships that could have appeared to influence this
467 work.

468 **Data and code availability**

469 The experiments use the public ADE and CoNLL04 distributions in
470 SpERT format. An anonymized reproducibility package will contain the

split construction, dataset statistics, source hashes, evaluation code, and derived result tables. Dataset text and model weights remain subject to their original licenses. The identifying repository URL is withheld during double-blind review.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used OpenAI Codex to assist with code inspection, language editing, figure preparation, and manuscript organization. The authors are responsible for reviewing the resulting material and for the final content. The language models evaluated as research systems are described separately in the Methods.

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